

Homework 2: AI Content Monitoring and Generation Workflow

Objective

The objective of this assignment is to design and document an **AI-powered workflow and agentic system** that continuously monitors, filters, classifies, and synthesizes the latest developments in Artificial Intelligence (AI) and its business applications. The final output should generate **high-quality, domain-relevant thought leadership content** suitable for professional platforms such as LinkedIn.

This assignment emphasizes **information prioritization, relevance routing, classification, and content generation**, within a real-world business decision-making context.

Assignment Requirements

1. Daily AI News and information Monitoring (15 points)

Design an AI agent that **checks the latest AI-related news and information on a daily basis**. You may consider store these information and insights in a repository using a vector database or regular database.

The following sources are recommended, but not limited to :

1. Venture Capital Firms

- Reports and insights from leading VCs such as:
 - Y Combinator
 - Andreessen Horowitz (a16z)
 - Sequoia Capital
 - Other top global venture capital firms

2. Investment Banks and Financial Institutions

- Research reports and insights from:
 - Goldman Sachs
 - JPMorgan
 - Morgan Stanley
 - Other major investment and asset management firms

3. AI Thought Leaders and Key Opinion Leaders (KOLs)

- Recent talks, interviews, technical blogs, or social media insights from:
 - Andrew Ng
 - Andrej Karpathy
 - Jensen Huang
 - Demis Hassabis
 - Mustafa Suleyman
 - Kai-Fu Lee
 - Lex Fridman

4. Top Business Schools and Academic Publications

- Insights and articles from:
 - Harvard Business Review
 - MIT Technology Review
 - MIT Sloan Management Review
 - Stanford University
 - Other leading global business schools and academic institutions

5. Management Consulting Firms

- Reports and thought leadership from top consulting companies (e.g., McKinsey, BCG, Bain, Accenture, Deloitte, etc.)

6. Leading Technology Companies

- Research blogs, product announcements, and white papers from:
 - Google
 - Microsoft
 - OpenAI
 - Other major AI and technology companies
-

2. Relevance Routing and Domain Definition (15 points)

Build a **relevance router** that evaluates whether each piece of information is relevant to:

- **Your professional role**, and
- **Your selected industry and business domain** (e.g., finance, consulting, healthcare, retail, manufacturing, SaaS, etc.)

You must:

- Clearly define your **industry, domain, and target audience**
 - Choose one or some of the potential relevance criteria, such as:
 - Strategic impact
 - Revenue or cost implications
 - Competitive advantage
 - Operational efficiency
 - Regulatory or governance considerations
-

3. Information Classification (15 points)

For all information, news or documents deemed relevant, design and implement an **AI-based classification mechanism** that groups information into several meaningful, business-relevant categories.

Example categories (you may define your own):

- Strategy and Executive Decision-Making
 - Product and Service Innovation
 - Infrastructure, Models, and Platforms
 - Governance, Ethics, and Regulation
 - Industry-Specific AI Use Cases
-

4. LinkedIn content research (15 points)

Choose 2–5 KOLs (Key Opinion Leaders) you admire, reverse engineer their content Document:

- How they **open** (hooks)
- How they **structure** (scannable formatting)
- How they **build credibility** (data, cases, opinions)
- How they **invite engagement** (questions, provocations)
- Their style: concise vs narrative, contrarian vs instructional, etc.

Output: a short “LinkedIn post style and anatomy” checklist that you’ll embed into your

workflow as a content generation guideline and evaluation rubric.

5. LinkedIn Content Generation (20 points)

Choose 2-3 categories in step 3. For each category:

- Generate **professional LinkedIn contents** with both text and image using the guideline in step 4:
 - Translates AI developments into clear business implications
 - Demonstrates depth of analysis and strategic thinking
 - Is concise, engaging, and suitable for executives, founders, or investors

You must specify:

- Target audience for each post
 - Tone and positioning (e.g., thought leadership, insight-sharing, executive briefing)
-

6. Workflow, Prompt Optimization and Progress Report (20 points)

Prepare a structured **progress report** documenting:

- Workflow architecture and design logic
 - Key challenges encountered
 - Workflow and prompt optimization progress, explaining What was improved, How the changes enhanced content quality, efficiency, or consistency etc
 - Lessons learned
 - Future opportunities for improvement or automation
-

Final Deliverables

You must submit the following:

1. **Workflow files**
 2. **Prompt Design Samples**
 - Key prompts used at each stage of the workflow
 3. **Final LinkedIn Content**
 - One post per category
 4. **Progress Report**
-

Evaluation Criteria

- Soundness of workflow and agent design
 - Effectiveness of relevance routing
 - Quality and business value of insight classification
 - Clarity, relevance, and professionalism of LinkedIn content
 - Depth of reflection and continuous improvement
-